
Phase 8 – Business Problem Solving

Project: Enterprise ERP Analytics Implementation

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1. Inventory increase despite flat sales

Analysis: Inventory (Closing Qty) shows a continuous increase over time, while sales (Order Qty) remain relatively constant.

Insight: Inventory has increased by ~20–30% while sales growth is stagnant, indicating inefficient inventory planning.

Suggestion :

Reduce procurement volume.

Implement demand forecasting models.

Introduce inventory control policies (EOQ).

2. High revenue but low margin products

Analysis: Some products generate high revenue but have low profit margins due to high procurement or logistics costs.

Insight: Top-selling products contribute significantly to revenue but yield poor profitability.

Suggestion :

Optimize pricing strategy.

Negotiate supplier costs .

Reduce logistics expenses.

3. Warehouse capacity utilization

Analysis: Warehouse stock levels vary significantly, with some warehouses nearing full capacity while others are underutilized.

Insight: Uneven distribution of inventory leads to inefficiency and increased holding costs.

Suggestion :

Redistribute inventory across warehouses.

Optimize warehouse allocation strategy.

4. Supplier total cost of ownership

Analysis: Supplier evaluation based on purchase cost, delivery delay, and logistics cost.

Insight: Low-cost suppliers often have high delays, increasing total operational cost.

Suggestion :

Select suppliers based on total cost, not just purchase price.

Prioritize reliable suppliers.

5. Fraudulent or suspicious transactions

Analysis: Certain transactions show unusually high order values or irregular patterns.

Insight: Outliers and duplicate patterns indicate potential fraud or data anomalies.

Suggestion :

Implement transaction validation rules.

Monitor high-value transactions.

Use anomaly detection models.

6. Costliest logistics routes

Analysis: Logistics cost varies significantly across different routes and carriers.

Insight: Some routes consistently incur higher transportation costs.

Suggestion :

Optimize routing strategies.

Select cost-efficient carriers .

Consolidate shipments.

7. Delayed deliveries & SLA failures

Analysis: Delivery time analysis shows frequent delays beyond expected timelines.

Insight: Certain couriers consistently underperform, causing SLA violations.

Suggestion :

Replace underperforming couriers.

Introduce SLA monitoring dashboards.

Penalize delays contractually .

8. Premium customer identification

Analysis: Customers are segmented based on revenue contribution and order frequency.

Insight: A small percentage of customers contribute the majority of revenue (Pareto principle).

Suggestion :

Introduce loyalty programs .

Provide personalized offers .

Prioritize premium customers .

9. Inventory liquidation opportunities

Analysis: Products with high stock levels but low demand are identified.

Insight: Dead stock occupies warehouse space and increases holding costs.

Suggestion :

Offer discounts / clearance sales.

Bundle slow-moving products .

Stop reordering low-demand items.

10. Logistics cost optimization

Analysis: Comparison of cost vs distance and carrier efficiency.

Insight: Some carriers charge higher costs for similar distances.

Suggestion :

Switch to cost-efficient carriers.

Optimize route planning.

Negotiate freight contracts.

11. Procurement strategy improvements

Analysis: Supplier performance analyzed based on cost, reliability, and lead time.

Insight: Not all low-cost suppliers are efficient due to delays and inconsistencies.

Suggestion :

Develop strategic supplier partnerships.

Use multi-supplier sourcing.

Implement vendor performance tracking.

12. Highest business-risk KPIs

Analysis: Key risk indicators include:

Stock-out rate

Delivery delay rate

High logistics cost

Insight: High delays and stock-outs directly impact customer satisfaction and revenue.

Suggestion :

Monitor KPIs through dashboards

Set threshold alerts

Take proactive actions

FINAL CONCLUSION

The analysis highlights inefficiencies in inventory management, supplier performance, and logistics operations. By leveraging data-driven insights, the organization can optimize costs, improve service levels, and enhance overall business performance.

This analysis demonstrates the use of data analytics in solving real-world supply chain and logistics challenges, enabling informed decision-making and operational efficiency.